4.3.2 Variation of atomic and ionic size:

It is generally expected a steady decrease in atomic radius along a period as the nuclear charge increases and the extra electrons are added to the same sub shell. But for the 3d transition elements, the expected decrease in atomic radius is observed from Sc to V , thereafter up to Cu the atomic radius nearly remains the same. As we move from Sc toZn in 3d series the extra electrons are added to the 3d orbitals, the added 3d electrons only partially shield the increased nuclear charge and hence the effective nuclear charge increases slightly. However, the extra electrons added to the 3d sub shell strongly repel the 4s electrons and these two forces are operated in opposite direction and as they tend to balance each other, it leads to constancy in atomic radii.

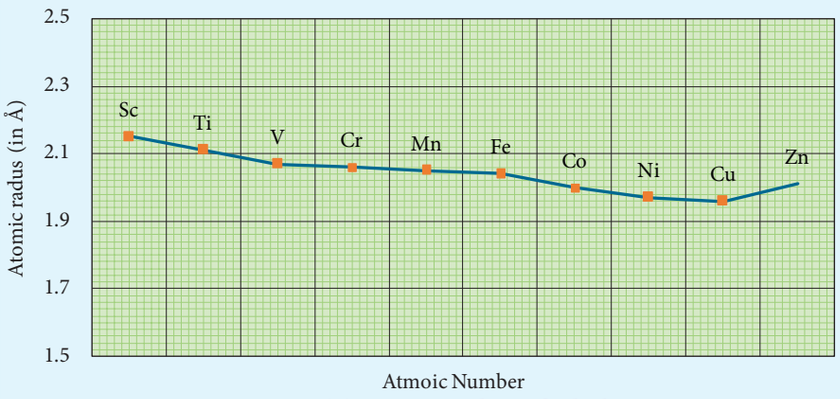


Figure 4.4 (a) Atomic radius of 3d Elements

At the end of the series, d – orbitals of Zinc contain 10 electrons in which the repulsive interaction between the electrons is more than the effective nuclear charge and hence, the orbitals slightly expand and atomic radius slightly increases.

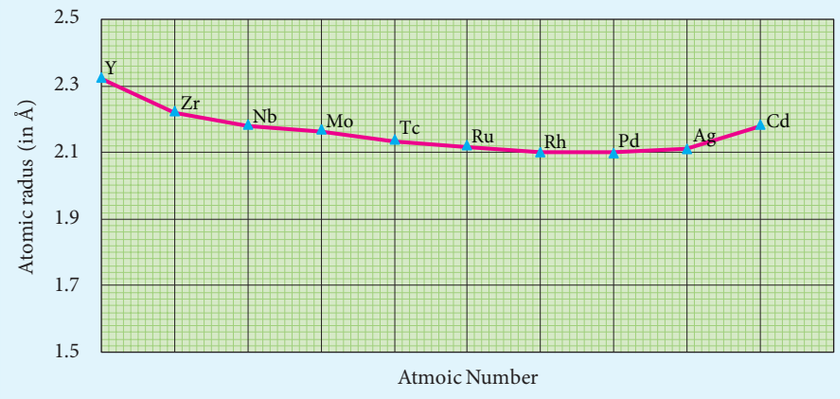


Figure 4.4 (b) Atomic radius of 4d Elements

Generally as we move down a group atomic radius increases, the same trend is expected in d block elements also. As the electrons are added to the 4d sub shell, the atomic radii of the 4d elements are higher than the corresponding elements of the 3d series. However there is an unexpected observation in the atomic radius of 5d

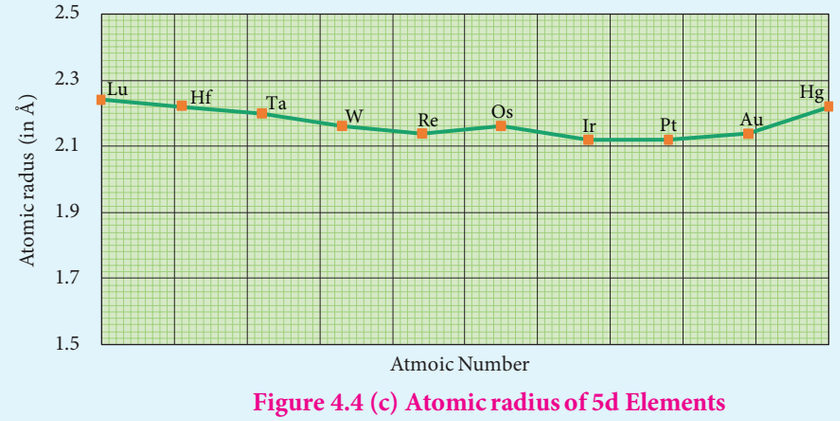


Figure 4.4 (c) Atomic radius of 5d Elements

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